

Gaps in the Rock Record — Teacher Guide

One HTML file, offline, no AI at runtime. You need: a screen to project or share; no other materials.

Learning goals

Not stated objectives — what the activity itself demonstrably shows, one per stage:

- 1. A local last observation in sampled rock is where the record stops, not a proven extinction moment.
- 2. Erosion deletes rock and its fossils outright; nothing is relocated to the boundary that erosion leaves behind.
- 3. A second section can show what a gap hides locally, but two local sections still cannot certify a global claim.

Where this sits on the path

This is one of ten Part-four example projects: an AI-built classroom tool whose goal is a bounded rock-record inference, whose engine is a live synthetic ledger computed as you sample, and whose assumptions — no dating uncertainty, no global distribution model, no fossil relocation — are one tap away in Details. It does not build on a specific earlier demo in the tour; it applies the tour's own discipline (name the goal, expose the assumptions, keep the evidence available) to a rock column instead of a model.

One question for the session

How much time does the rock in front of you actually show, and how much is missing? Surviving rock and elapsed time are different measurements, and this activity never lets you compute one from the other by eye — only supplied chronology evidence fixes an exact Myr value.

*The one sentence to land: **a local rock record can only ever tell you where ITS evidence stops** — not where the organism, the deposition, or the story actually ended.*

The shape of the session

STAGE	WHAT HAPPENS	TIME
1 — Section A	A continuous three-layer record. Sample every layer, reveal supplied ages, and test whether a fossil's last appearance settles extinction.	~3 min
2 — Section B	An erosional gap. Reveal the missing interval, sample above and below it, and test whether the gap explains its own cause.	~3 min
3 — Sections C & D	A paired comparison: C pauses, D keeps depositing. Sample both to test what a second section can and cannot settle.	~3–4 min

Stage 1 — Section A

Question. Does a species' last appearance in sampled rock mark where it went extinct?

Predict prompt. Before sampling, guess: does finding where a fossil stops appearing settle whether the species went extinct there?

Activity. Sample all three layers in Section A (Blue mudstone, Green mudstone, Gold sandstone), then press **Show supplied age evidence**. *Asteria* appears in the lower two layers and nowhere above; ask the room what that absence does and does not prove before choosing a conclusion.

Expected observation. Every layer samples cleanly with no gap in this section (the evidence ledger and the linked time view both confirm a continuous 6 Myr record); the "warranted" conclusion is available only after middle and top are sampled and the chronology is shown.

Check-yourself answer key. "Does *Asteria*'s absence above Blue and Green mudstone settle global extinction?" — No: a local last observation is not a global extinction record.

Teaching limit. This section has no missing interval at all; it is here to establish the extinction-inference limit before Stage 2 adds a genuine gap.

Stage 2 — Section B

Question. Is a rock gap always someone's last moment, and do a gap's missing fossils end up sitting at its boundary?

Predict prompt. Before looking, guess: once you find rock missing here, will the surviving evidence also tell you why it is missing?

Activity. Press **Show supplied age evidence** first so the gap is visible, then sample Red siltstone (below) and Cross-bedded sand (above). Reveal the authored history and have the room compare "what the gap alone showed" against "what erosion actually did."

Expected observation. The reveal states the erosion event removed 15.00 m of Pale limestone entirely and 10.00 m from the top of Red siltstone, taking their fossils with it — nothing from the removed rock reappears at today's contact.

Check-yourself answer key. "Are the removed fossils now sitting at the contact?" — No: erosion deletes rock and fossils; nothing relocates to the boundary.

Teaching limit. This section's gap happens to be erosional; a gap does not require erosion in general (Section C's pause in Stage 3 has none at all). Do not let the room generalize "every gap is erosion" from this one case.

Stage 3 — Sections C and D

Question. Does a second section that keeps depositing through a first section's gap settle what the gap means?

Predict prompt. Before comparing, guess: if Section D preserves fossils from exactly the span missing at C, does that settle the question for the species everywhere?

Activity. Reveal supplied chronology for both C and D, then sample below C's gap and the matching Green mudstone layer in D. The evidence-readiness check requires samples and chronology from BOTH sections before offering a conclusion — point this out as a deliberate gate, not a bug.

Expected observation. D's Green mudstone preserves a fossil from exactly the 2.00 Myr span that C's pause leaves blank; the room should see this as "D fills in what C cannot show," not as proof that the species was present everywhere during that span.

Check-yourself answer key. "Do C and D together prove a global claim?" — No: two local sections are not a global sample, however informative.

Teaching limit. The two sections are given identical starting rock so the pause is the only variable; a real paired-section study would need to establish correlation between sections independently, which this activity does not model.

Misconceptions

"Once a fossil stops appearing, the organism is extinct."

No. A local last observation is where a section's own evidence stops, not where the organism itself ended everywhere (Stage 1).

"Equal rock thickness means equal elapsed time."

No. Section A's own three layers deposit at different rates (10, 7 and 8 m/Myr) over the identical 2 Myr each, so they come out different thicknesses for the same elapsed time (Stage 1).

"A rock gap always means erosion happened."

No. Section B's gap is erosional, but Section C's pause in Stage 3 is a gap with no erosion at all — non-deposition is a separate, equally valid cause (Stages 2–3).

"Two sections that compare well together certify a global claim."

No. Comparing Sections C and D is more evidence than either alone, but it is still two named local places, not a global or basin-scale survey (Stage 3).

Discussion questions

- Section A's three layers each took 2 Myr to deposit but came out different thicknesses. What would go wrong if you tried to read elapsed time directly off a ruler held against the rock?
- Section B's contact would look the same on the surviving rock whether the missing time came from erosion or from a long pause. What additional, real-world evidence would let a geologist tell those two apart, that this activity does not model?
- If Section D's rock at the matching interval had ALSO been missing, would that change what you could conclude about Section C's gap? Why or why not?

Questions you will get

"Doesn't the app already know the real history? Why hide it?"

It does — every case has an authored reveal. The point of hiding it until a conclusion is chosen is to force the claim to rest on the VISIBLE evidence, not on knowing the answer key. That is also why choosing a conclusion is not scored: it is feedback on the claim, not a test of memorised history.

“Why can’t I just measure the rock and compute the missing time myself?”

Because that is the specific habit this activity is built to interrupt. The only source of an exact Myr value anywhere on screen is **Show supplied age evidence** — a stand-in for independent chronological evidence (radiometric dates, correlated marker beds) that a real geologist would need. Nothing here lets you derive an exact age from thickness and an assumed rate.

“Is this how real geologic gaps work?”

Only in outline. This is a small authored teaching fixture: it omits reworking, transport, dating uncertainty, sampling statistics, facies correlation and any global distribution model. Each stage's Details drawer states its own boundary plainly — open one in front of the room if this question comes up.

Presenter controls

- Turn on **Settings** → **Presentation mode** for projector-sized type.
- Between rooms, **Settings** → **Reset** puts the demo back to its start **in place** — **no page reload**: every guess and sample cleared in all three sections, supplied age evidence and revealed histories hidden again, every conclusion unchosen, and Section A selected. Presentation mode stays on so the projector does not change under you.
- The app saves nothing anywhere, so an ordinary page reload resets it too (a reload is simply a more disruptive way to reach the same state Reset reaches in place).
- No network is needed at any point. If the room's wifi is down, nothing changes.

The buttons, top right

- ? — **Guide**: the short how-to panel that opens on first load, reopenable at any point. (Not this document.)
 - ⚙ — **Settings**: **Open Presenter Notes** (this document, on screen), **Presentation mode**, and **Reset**.
-

Gaps in the Rock Record — Bryant Harrison, Murray State University. This is a small authored teaching model, not a basin simulator or a method for establishing real extinction. Its event/occurrence framing was developed with reference to the StratPal event-data tutorial; this demo preserves authored synthetic specimens rather than treating occurrence-event transformations as a specimen-preservation algorithm. No faculty endorsement is claimed.